Yucheng (Cecilia) Zhang

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Education

Northwestern University, M.S. Computer Science — Sep. 2026 – May 2028, Evanston, IL
The University of Texas at Austin, B.S. Mathematics; CS Minor; GPA: 3.74/4.00 — Aug. 2023 – May 2026, Austin, TX

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, R
Backend/Web: React, Flask, Node.js, REST API development, SQLAlchemy, Alembic, Pytest, Jest
Data/Cloud: PostgreSQL, SQL Server, Azure Synapse, Databricks, Docker, Git, Power BI
AI Integration: OpenAI, Gemini, Claude, LLM APIs, prompt configuration, context filtering, multi-provider routing

Experience

Enterprise Risk Analyst Intern — May 2025 – Dec. 2025

Toyota Industries Commercial Finance | Python, SQL, Azure, Power BI — Dallas, TX

- Automated monthly credit-risk model refreshes with parameterized SQL and Python workflows connected to Power BI, reducing manual reporting effort by 10%.
- Built ETL and reconciliation logic across PayNet and internal loan servicing data, standardizing features including inquiries, LTV ratio, and account tenure for recurring risk model refreshes.
- Delivered Azure and Power BI dashboards for recurring risk monitoring across Class 8 trucks, trailers, and used-vehicle loan portfolios.
- Validated Logistic Regression and Random Forest outputs with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** and improving high-risk case identification by **3–4%**.

Financial Data Development Engineer – Fraud Detection — Jul. 2024 – Aug. 2024

Caitong Securities Ltd | SQL, Oracle, Stored Procedures — Hangzhou, China

- Designed SQL ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Built modular fraud-detection logic and a customer risk-scoring engine from transactional and behavioral streams, reducing false positives by 2% in daily monitoring workflows.
- Organized stored-procedure logic into reusable modules so risk rules and threshold updates could be maintained without rewriting the full transaction pipeline.

Software Engineering Intern – Customer Analytics — Nov. 2023 – Jun. 2024

Joblogic-X Corporation | Python, APIs, Git — Dallas, TX

- Engineered a pipeline to scrape and process XHS social-media data and join it with transactional order systems for repeatable customer analytics storage.
- Implemented modular Python ranking and clustering algorithms, generating reproducible customer-priority lists for marketing and engagement workflows.
- Integrated a KOC scoring module into Meet Fresh's customer-facing chatbot through REST APIs, enabling ranked recommendations inside the existing workflow.
- Maintained Git-based deployment workflows to keep customer analytics updates reproducible and reviewable across local and remote development environments.

Projects

CodeAssist – Full-Stack AI Feedback System | GitHub — Aug. 2025 – Present

University of Texas at Austin | Faculty Project | React, Flask, PostgreSQL, Docker — Austin, TX

- Built a full-stack AI feedback system for programming courses, connecting student code editing, file uploads, submissions, and chat-based AI feedback for debugging, complexity reasoning, and test-case correctness.
- Developed Flask REST APIs and PostgreSQL models for AI feedback requests, instructor settings, student submissions, backend validation, JSON serialization, and Alembic migrations, keeping schema changes reproducible.
- Implemented context-filtering logic so assignment descriptions, student code, uploaded files, tests, results, and output are sent to AI providers only when instructor policies allow them, reducing data oversharing risk.
- Standardized provider and assignment-level settings across OpenAI, Gemini, Claude, and local models, including model selection, API configuration, timeout handling, and role permissions.
- Built local stress-test runners with server health checks, structured JSON reports, cleanup controls, and documented enrollment, submission, grading, and CRUD workflows.

Echo – Interactive AI Voice Simulation System — Jun. 2026 – Present

University of Texas at Austin | Mentor: Prof. Bo Xie | Next.js, TypeScript, OpenAI API — Austin, TX

- Built an event-driven Next.js app that generated context-aware voice responses from pauses, errors, skipped fields, and task switching using session state, browser TTS, and OpenAI APIs.
- Designed cooldown, interval, and event-priority logic to prevent repetitive playback while keeping generated responses tied to the current task state.
- Modularized prompt templates and event-trigger logic so new voice events and task types could be added without changing the core session controller as requirements changed.